

Link-Aware Shrinking and Selective Prefetch in Heterogeneous Wireless Clusters: Benefits and Limits

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Abstract—We evaluate shrinking compute- and link-aware chunks, stationary-only prefetch, a completion cutoff, and fenced task recovery in heterogeneous wireless clusters. A reproducible SimPy reconstruction combines paired benchmarks, mechanism ablations, calibrated simple comparators, a restricted HEFT comparator, and censoring diagnostics. An independent ns-3 radio diagnostic supplies a bounded alternative mapping. Against the calibration-selected fixed-size baseline on untouched combined-cell seeds, signed improvement is 6.26% (paired restricted-time difference interval [90.93, 194.84] s). Stationary-only prefetch underperforms unrestricted prefetch in the nominal combined cell by a signed -5.75%. With the independent ns-3-derived mapping in a stationary mixed-link cell, improvement over the fixed-size comparator is 0.28% (interval [0.04, 7.95] s). All comparative inference uses the expanded global family; the ns-3 diagnostic does not validate end-to-end mobile scheduling. The results support a study of conditional benefits and mechanism failures. Physical mobile scheduling, faithful close mobile-framework comparison, and full-text novelty verification remain outstanding.

Index Terms—heterogeneous scheduling; wireless workers; guided chunks; selective prefetch; shared channel; task recovery; reproducible simulation.

I. INTRODUCTION

Nearby computers can contribute processing capacity without an Internet connection. However, scheduling by processor speed alone can overassign a device whose inputs arrive slowly, while equal-sized allocations can leave a slow worker holding the tail. Prefetch can overlap communication and computation, but prefetching to a mobile device may reserve work that must later be repeated. These tradeoffs motivate a coordinator that distinguishes stationary and mobile participation and reduces its allocation granularity as the job progresses.

The central question is not whether shrinking chunks or prefetch can be invented anew. Both have substantial prior art. The question is whether their constrained combination with observed link quality provides a reproducible incremental benefit over strong simple policies, and under which conditions that benefit disappears. The contribution examined here is a particular combination and its mechanism-level evaluation, rather than a universal winning scheduler.

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The study provides an executable fluid-channel SimPy model; matched mechanism ablations and classical scheduling analogues; censored-run-aware paired comparisons; and explicit recovery-counter tests. It does not demonstrate a real wireless-cluster speedup. The proposed completion cutoff is a fixed user-specified design setting, not an empirically established optimum.

II. RELATED WORK AND COMPARISON SCOPE

Guided self-scheduling and factoring [1], [2] motivate allocating a decreasing share of remaining tasks. Weighted variants account for unequal processors. LB4OMP [3] and MPI self-scheduling implementations establish important algorithm families, but their runtime assumptions differ from a shared wireless master-worker system. The simulated chunk rules used here are defined mathematically and are not represented as complete implementations of those runtimes.

MAESTRO [4], CROWDio and dependency-aware Honeybee frameworks [5] are close systems-level comparisons because they combine capability-aware allocation, nearby mobile resources or recovery. The supplied synopsis identifies these studies and Forseti [6], adaptive work stealing, and speculative straggler mitigation as related work. A faithful comparison requires their available code, workload assumptions and communication interfaces. The present artefact therefore includes a clearly labelled speculative-copy adaptation rather than relabelling it LATE, Dolly or a crowd-computing framework.

Reference audit and scope. Crossref metadata now verifies all eleven retained references, including the original seven; the artefact retains the metadata responses. Metadata verification does not establish that every implementation reproduces a published system. The HEFT ranking and earliest-finish core [7] is additionally evaluated on independent tasks with stationary workers. Its oracle initial CPU speeds favour this offline comparator, while shared-radio execution remains an adaptation. A faithful close mobile-framework comparison remains outstanding.

The 802.11 performance anomaly [8] motivates sensitivity to airtime versus throughput fairness. The independent packet-level diagnostic uses ns-3 [9]. ADWIN [10] appears only in the optional detector implementation and its tests; the main scheduler excludes it.

III. SYSTEM MODEL

A wired coordinator manages independent, indivisible task identifiers. Each worker computes locally and exchanges in-

put and result data over a shared fluid channel. The world contains task costs, input sizes, compute-speed traces, worker positions, radio observations and availability. A deterministic hash of scenario parameters and the evaluation seed initializes all world randomness. Scheduling policies draw no random numbers. Changing the administrative horizon does not change the physical trace prefix.

Task cost is measured in reference-worker compute seconds. Default inputs are proportional to cost; robustness cells use heavier-tailed costs, independent input sizes and a larger result/input ratio. Compute heterogeneity preserves total nominal speed while changing its distribution. Permanent workers retain their initial coordinates throughout execution. Dynamic workers may move, experience a temporary dip, permanently leave, or leave and return. Signal strength can fluctuate even at a fixed position.

A. Radio and delay assumptions

Received signal strength is

$$\text{RSSI}_i(t) = P_0 - 10n \log_{10} \left(\frac{d_i(t)}{d_0} \right) - L_{\text{wall}} \mathbf{1}_{\text{wall}} + \eta_i(t) - \Delta_i(t). \quad (1)$$

The implemented parameters, throughput interpolation knots and shadowing assumptions are recorded in the source and experiment configuration. The interpolation knots are reconstruction assumptions, not measurements recovered from the missing original artefact. Core cells deliberately override geometric RSSI with controlled strong and weak link groups to isolate mechanisms; moving coordinates alone do not generate their disconnections, which are separately seeded availability events. Default RTT is an exponential function of RSSI; a separate robustness cell adds delay independent of RSSI. Thus, the default RSSI and RTT terms contain correlated information. Each chunk includes a delay corresponding to its observed RTT before input transfer.

For transmitting transfers with positive table rate T_i , the equal-airtime rule allocates $r_i = T_i/A$. Its aggregate rate is the mean active table rate. A throughput-fair rule allocates $r_i = (\sum_j 1/T_j)^{-1}$ to each active transfer. A third rule reduces equal-airtime rates by concurrency overhead and a low-rate delivery factor. Silent transfers receive zero rate and are excluded from the current transmitting set. This avoids treating an absent device as a station that must receive throughput.

All rules operate per transfer, not per physical station; concurrent uplink and downlink exchanges on the same worker are not a faithful MAC model. The overhead rule is explicitly uncalibrated and must not be called a calibrated Bianchi model. Unlike a MAC-level analysis [11], no result in this draft validates 802.11 frame contention, capture, retransmission, aggregation or driver behaviour.

IV. SCHEDULING AND RECOVERY

Let R be the unassigned task count, U the currently eligible worker set, and $\hat{s}_i$ a compute-only EWMA estimate from completed tasks. A new chunk has size

$$k_i = \min \left(R, \max \left(1, \left\lceil \sigma R \frac{\hat{s}_i}{\sum_{j \in U} \hat{s}_j} \ell_i \delta_i \right\rceil \right) \right), \quad (2)$$

where ℓ_i and δ_i are clipped RSSI and RTT factors. Estimates start equal and are updated from compute duration excluding CPU-resource waiting and communication. A physical prototype should include measured dry-run initialization and its cost. The prefetch trigger uses instrumented worker progress: remaining nominal compute cost is reported after simulated compute intervals and divided by the current speed estimate. This idealized progress telemetry and compute-state reporting have no modelled message delay or cost. Removing compute weighting holds these estimates equal; removing RSSI or RTT leaves the other factor unchanged.

Idle workers have dispatch priority over prefetch reservations. Only physically stationary workers may receive the next chunk while their current chunk computes, and only one next chunk may be outstanding. Prefetch requires the estimated remaining compute interval to exceed estimated input-transfer duration plus a margin. No new prefetch starts once the number of unique accepted results reaches the user-specified completion cutoff. An existing valid reservation is not revoked merely because the cutoff is crossed. This is a mechanism to limit tail reservation, not a guarantee of lower tail idle time.

Probes classify a link using reachability, RSSI and RTT thresholds. Repeated unhealthy observations enter bounded grace; expiry invalidates the session and returns unaccepted task IDs to the pending set. A disconnected worker can continue computing locally, so its late result is a genuine stale-session case. Task-id deduplication ensures that a result from a valid session is accepted at most once. Returning radio connectivity is not equated with proof that a worker physically departed or returned.

The exploratory speculative baseline creates a whole-chunk duplicate on an idle worker only after an assignment exceeds its estimated compute duration by a specified factor. It has task-id deduplication. It is deliberately an adaptation, not a faithful implementation of a named published policy. A single indivisible long task remains a lower bound on completion time; shrinking task groups cannot remove that bound.

V. EXPERIMENTAL PROTOCOL

The core study uses 50 fresh paired seeds in each of the uniform, weak-link, churn and combined cells. The robustness study uses 20 different seeds for contention sensitivity, independent heavy-tailed input/cost behaviour and additional RTT. The tuned-baseline study uses 20 calibration seeds and 50 separate evaluation seeds per core cell, with 800 additional calibration runs. Minimum worst-cell mean relative regret selects 1 task(s) per fixed chunk and weighted share 0.1. The grid and complete selection records are retained with the artefact. Weak-link horizon diagnostics use 10 further seeds. Configurations and source hashes accompany the completed core and robustness batches; diagnostic scripts and complete settings record the supplementary experiments. Calibration seeds are excluded from evaluation, and the original manuscript's evaluation seeds are not used. These settings are exploratory and were not independently preregistered.

The simulation batches passed 32 tests before execution. The final gate contains 39 tests, including additional

interaction-analysis, HEFT and calibration-support checks. It checks deterministic immutable worlds, link-rate properties, CPU exclusivity, byte accounting, horizon censoring, stationary coordinates, prefetch eligibility and cutoff, independent RTT weighting, and constructed late-result recovery. Detector tests are diagnostic only: the main proposed policy excludes ADWIN, so detector effectiveness is not a claimed contribution.

A. Estimands, uncertainty and multiplicity

For algorithm A against comparator B , define

$$\text{improvement}(A, B) = 100 \frac{\overline{\min(T_B, H)} - \overline{\min(T_A, H)}}{\overline{\min(T_B, H)}}. \quad (3)$$

Positive improvement means the candidate is faster through the common administrative horizon H . Every original seed enters this primary analysis. Fresh weak-link and repeated combined-cell diagnostics extend the horizon without changing the physical world prefix and includes optional hysteresis. Noncompletion at a finite extended horizon is not proof of infinite livelock. For common administrative censoring, the empirical mean of truncated completion times is also the restricted survival-time area through H . It is a lower bound on unrestricted mean completion time, not an estimate of that unobserved tail. Completion rates are always reported separately.

Bootstrap resampling operates on paired seeds with 5000 resamples. A two-sided paired sign-flip test uses exact enumeration for small seed counts and a finite-simulation-corrected Monte Carlo procedure otherwise. Assumptions include exchangeability of the within-seed signs under the tested null. Exact test type, permutation count and raw p are retained in the inference appendix. Holm adjustment covers all 149 candidate comparisons across the complete core and robustness studies. A practical win requires global Holm-adjusted $p < 0.05$ and improvement of at least one percent. Completed-pair comparisons remain a labelled sensitivity in the machine-readable analysis, not the primary evidence.

A post-hoc add-one-in decomposition isolates link factors and prefetch against the matched compute-only baseline. Where all four factorial combinations are available, the paired interaction is $T_{\text{link}} + T_{\text{prefetch}} - T_{\text{base}} - T_{\text{full}}$; positive values denote superadditive saving. These tests join the same global family. This is a two-mechanism interaction analysis, not a complete interaction analysis of all mechanisms. Paired standardized differences use the standard deviation of paired differences. The precision appendix reports an approximate unadjusted detectable paired difference for a conventional power target; it does not promise adequate power after global multiplicity correction. Marginal survival curves describe completion distributions; this draft does not report an ordinary independent-sample log-rank test as if paired worlds were independent.

VI. RESULTS

A. Calibration-selected baseline comparisons

Uniform versus Tuned-fixed: improvement 0.74% (difference interval [3.88, 8.22] s; global Holm $p = 0.0149$), which does not meet the stated win rule.

Uniform versus Tuned-weighted: improvement 1.75% (difference interval [10.93, 18.02] s; global Holm $p = 0.0149$), which meets the stated win rule.

Weak links versus Tuned-fixed: improvement 11.25% (difference interval [209.83, 287.52] s; global Holm $p = 0.0149$), which meets the stated win rule.

Weak links versus Tuned-weighted: improvement 44.43% (difference interval [1451.22, 1690.31] s; global Holm $p = 0.0149$), which meets the stated win rule.

Churn versus Tuned-fixed: improvement -0.94% (difference interval [-13.14, -3.02] s; global Holm $p = 0.2031$), which does not meet the stated win rule.

Churn versus Tuned-weighted: improvement 1.58% (difference interval [7.51, 20.33] s; global Holm $p = 0.0216$), which meets the stated win rule.

Combined versus Tuned-fixed: improvement 6.26% (difference interval [90.93, 194.84] s; global Holm $p = 0.0149$), which meets the stated win rule.

Combined versus Tuned-weighted: improvement 38.01% (difference interval [1201.48, 1414.98] s; global Holm $p = 0.0149$), which meets the stated win rule.

B. Matched nominal mechanism comparisons

uniform: relative to the matched compute-only scheduler, the policy changes restricted makespan by 3.70% (baseline minus candidate 31.20 s; paired interval [25.90, 36.48] s; global Holm $p = 0.0149$), and meets the stated statistical and practical win rule.

weak_links: relative to the matched compute-only scheduler, the policy changes restricted makespan by 49.08% (baseline minus candidate 1949.26 s; paired interval [1882.56, 2008.79] s; global Holm $p = 0.0149$), and meets the stated statistical and practical win rule.

churn: relative to the matched compute-only scheduler, the policy changes restricted makespan by 7.50% (baseline minus candidate 70.27 s; paired interval [58.97, 81.68] s; global Holm $p = 0.0149$), and meets the stated statistical and practical win rule.

combined: relative to the matched compute-only scheduler, the policy changes restricted makespan by 40.71% (baseline minus candidate 1616.91 s; paired interval [1483.90, 1742.54] s; global Holm $p = 0.0149$), and meets the stated statistical and practical win rule.

Complete comparison tables appear in the supplementary material; all original comparisons, including negative results, are retained.

VII. INDEPENDENT RADIO CHECKS AND POST-HOC DIAGNOSTICS

A. Packet-level radio diagnostic

An independent ns-3 diagnostic contains 120 runs of stationary IEEE 802.11a downlink saturation and separate unloaded

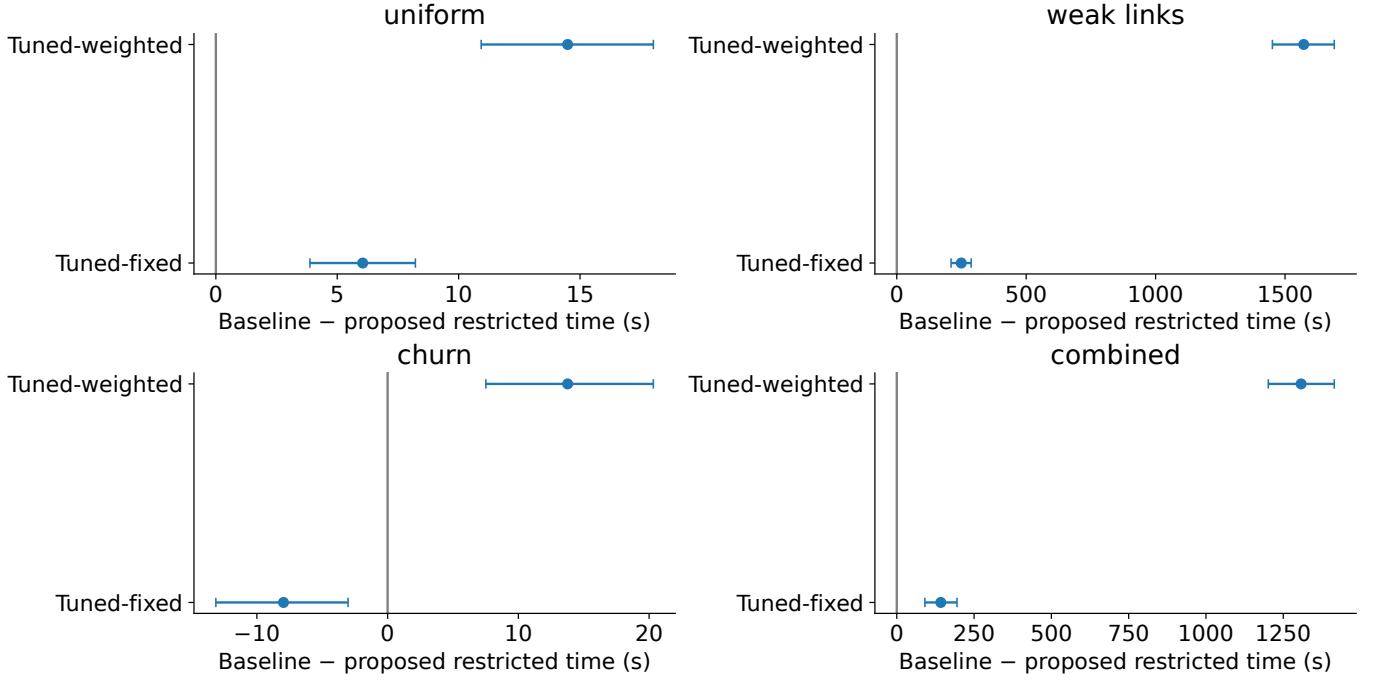


Fig. 1. Fresh-seed comparisons with fixed-size and weighted-shrinking baselines selected on separate calibration worlds. Negative differences favour the baseline.

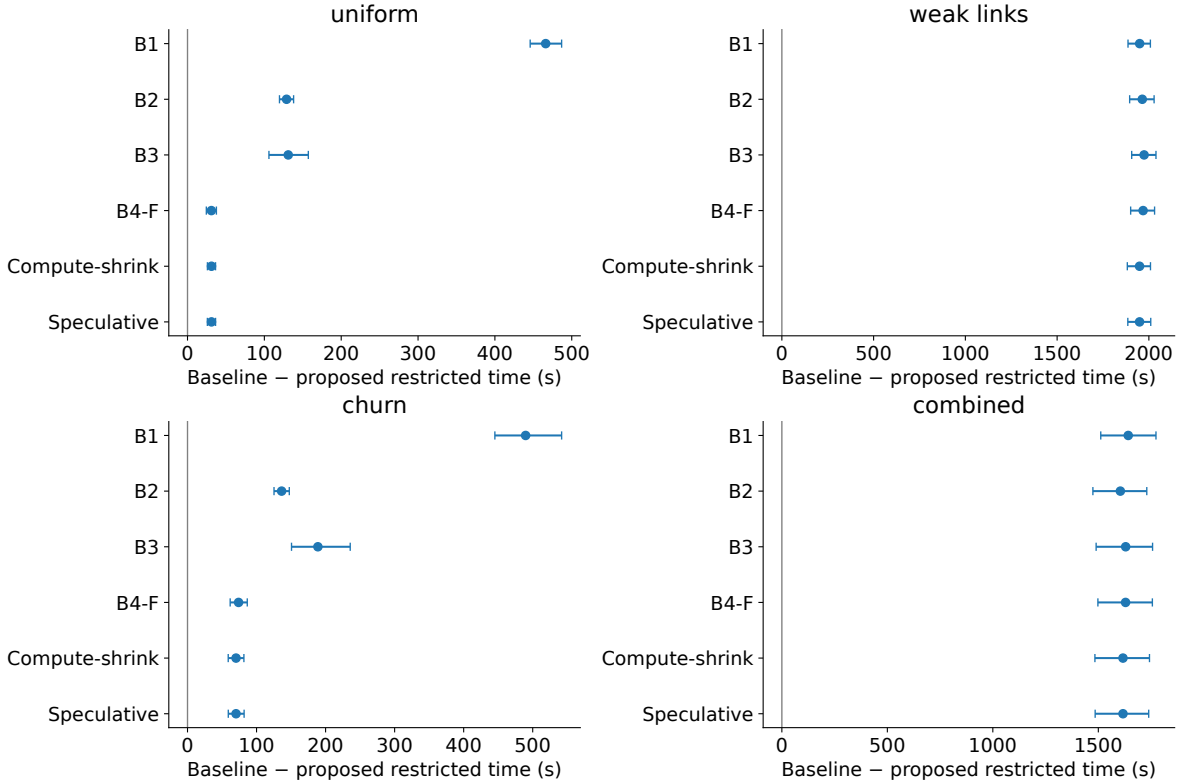


Fig. 2. Paired restricted-time differences against the nominated classical, matched and speculative baselines. Positive differences favour Proposed. Intervals resample paired seeds; all comparisons remain visible.

UDP echo trials. The model uses a single AP, ideal rate adaptation, log-distance loss without fading, co-located receivers at each distance, and no mobile scheduler execution. Single-

station application-goodput knots use RNG runs 1–3; holdout runs 4–5 test the same distances and additional receiver counts. This tests repeatability and concurrency generalization, not

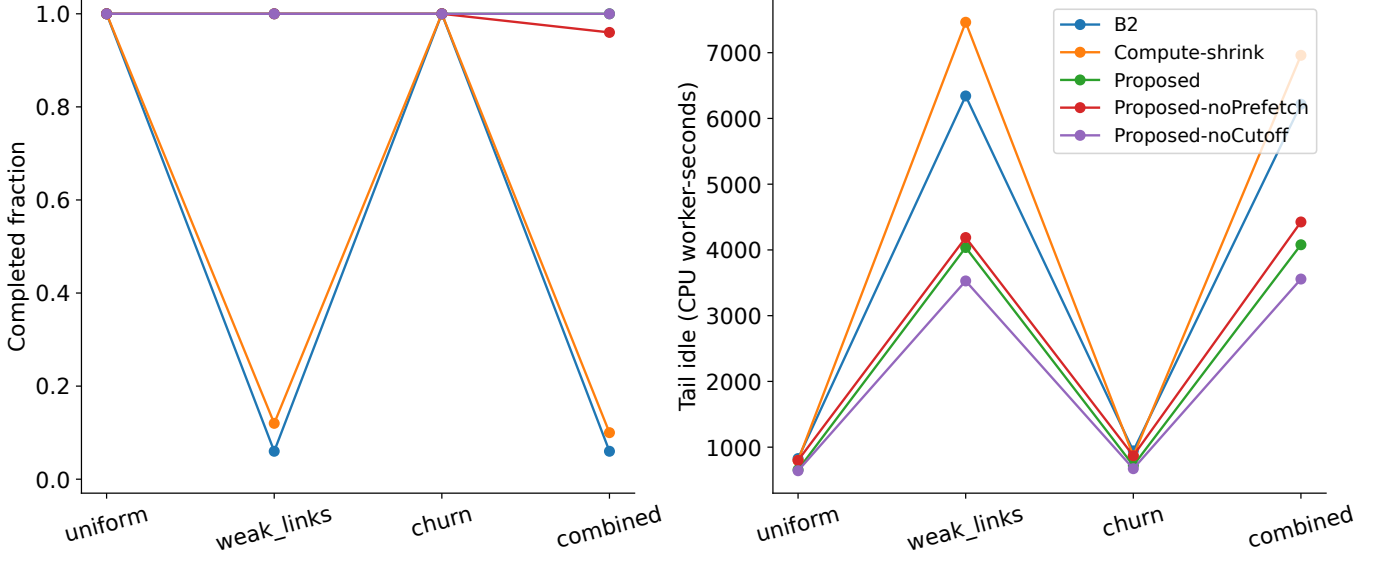


Fig. 3. Completion fractions and tail CPU-idle worker-seconds. The tail begins after the specified unique-result completion threshold. Disconnected workers remain in this nominal-capacity metric.

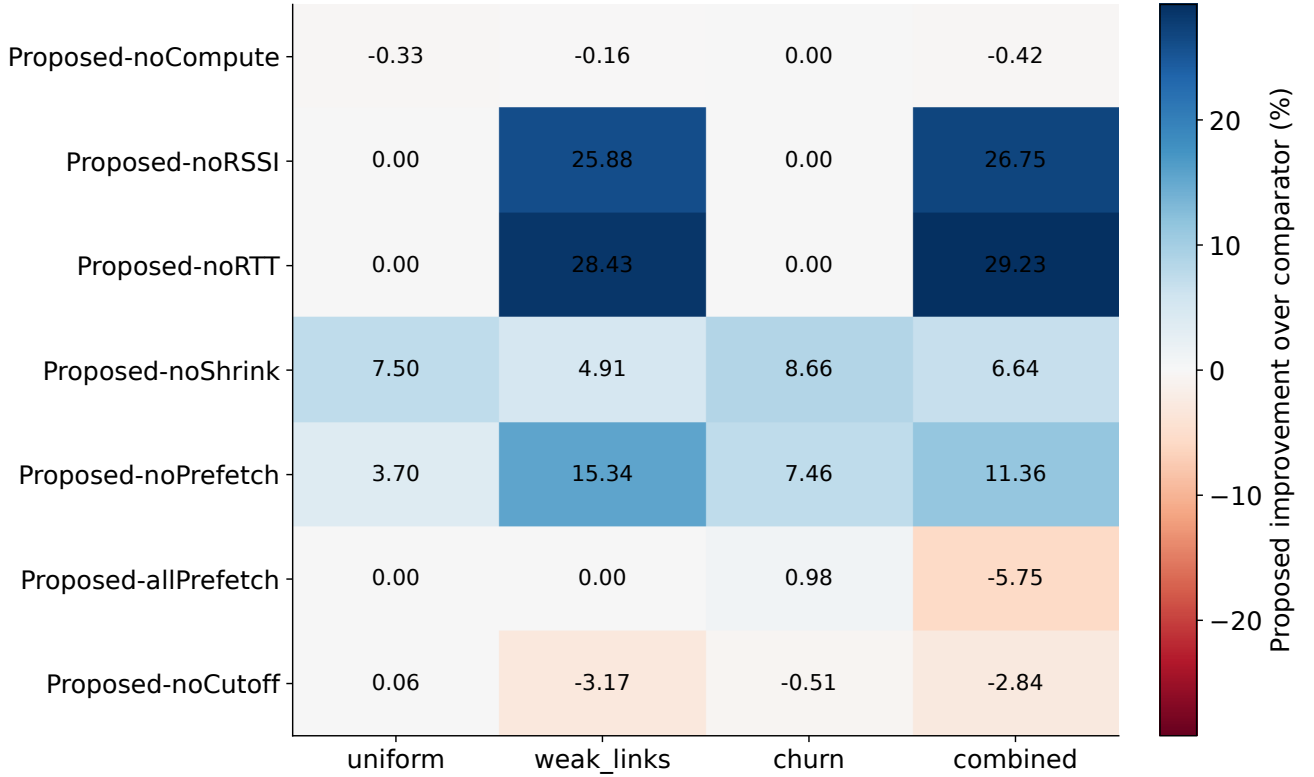


Fig. 4. Mechanism comparisons. Positive entries favour Proposed; a gain is not automatically statistically or practically significant. Generated comparison tables supply intervals and globally adjusted inference.

unseen-distance or physical-device accuracy.

Assumed-curve versus fitted-curve holdout aggregate-goodput MAE is 14.421 versus 0.027 Mbps for one receiver, 14.429 versus 0.017 Mbps for two, and 14.453 versus 0.030 Mbps for four. The very small fitted errors arise in a deliberately simple, unfaded topology and should not be generalized to real Wi-Fi. Unloaded echo averages 0.300 ms; the assumed

RSSI/RTT function has MAE 65.57 ms. Unloaded echo does not calibrate RTT under saturation.

B. Sensitivity to the ns-3-derived curve

Two additional stationary cells use fresh paired seeds, fixed strong or mixed RSSI values within the calibration support,

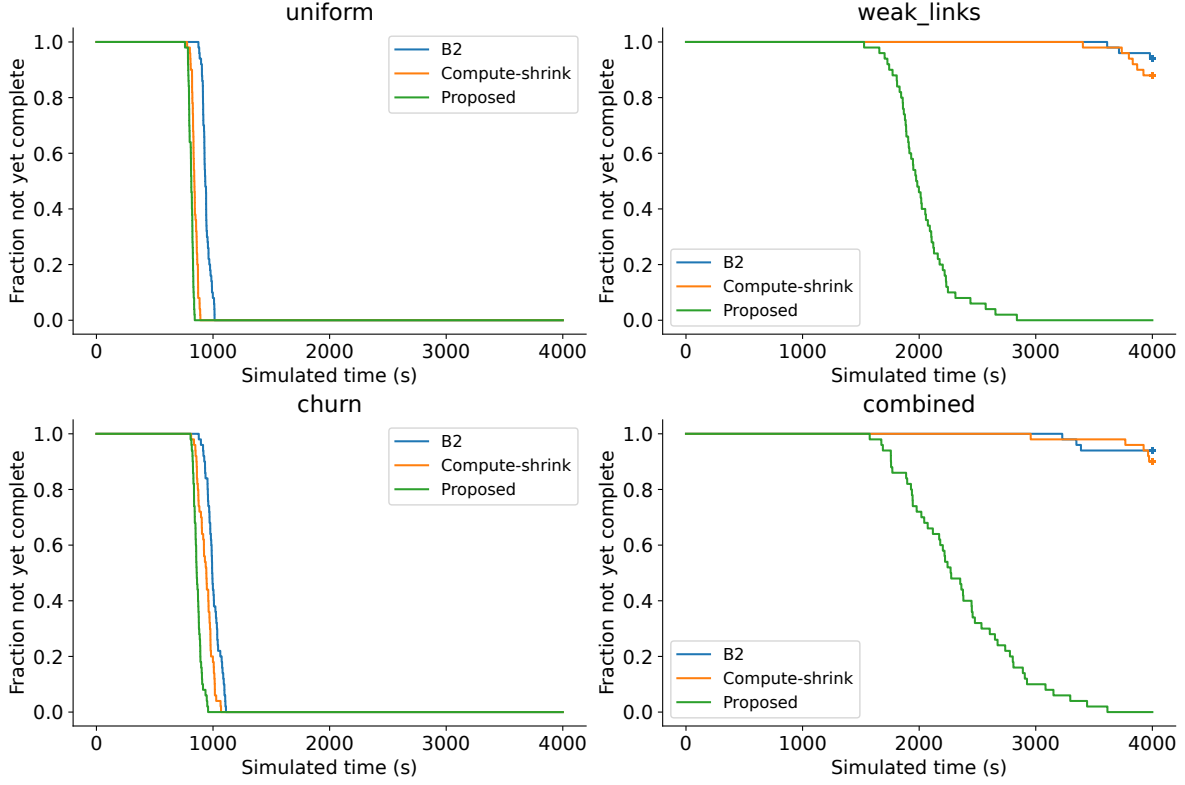


Fig. 5. Fraction of batches not yet complete, including administratively censored runs. Curves describe marginal completion behaviour; paired restricted-time inference is reported separately.

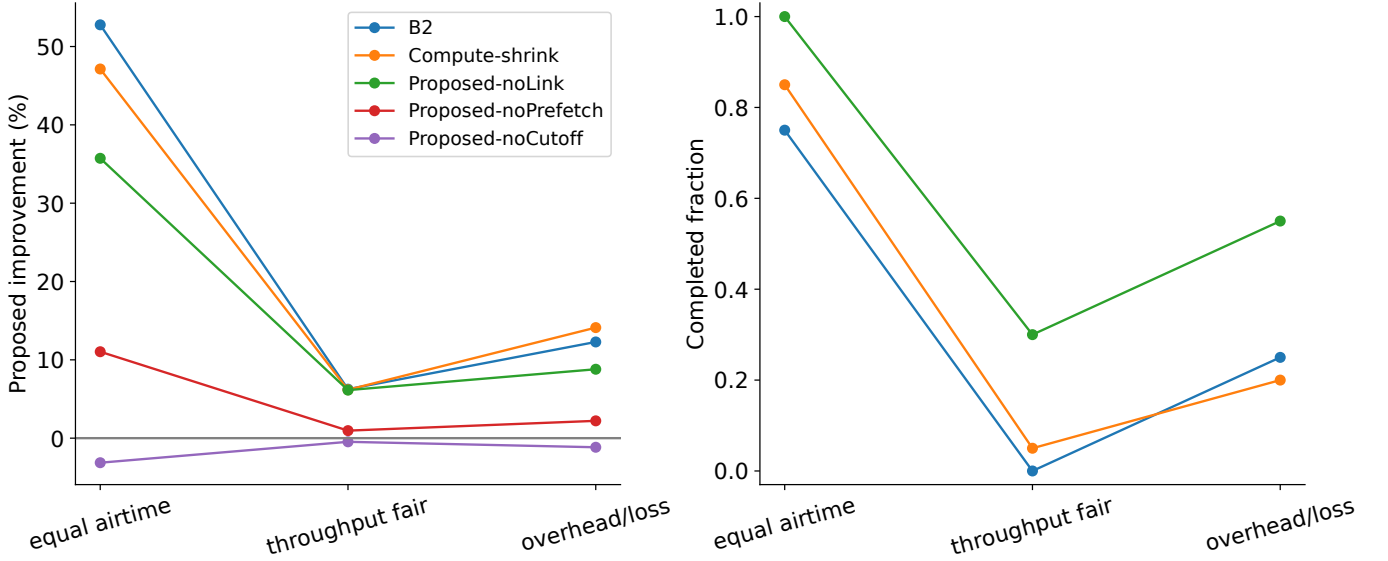


Fig. 6. Effect and completion sensitivity across alternative fluid contention assumptions. The overhead/loss model is not calibrated to measurements.

and no fading or departures. SimPy uses the training goodput knots and unloaded-echo RTT knots; no extrapolation is permitted. All original assumed-channel results remain retained. These experiments probe whether mechanism conclusions survive one alternative mapping, not whether complete scheduling gains reproduce in ns-3. Their comparisons join the expanded global Holm family.

Ns3 stationary strong versus Tuned-fixed: signed gain 0.06% (paired difference interval [-3.17, 4.26] s; global Holm $p = 1.0000$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary strong versus Compute-shrink: signed gain 2.05% (paired difference interval [15.13, 34.76] s; global

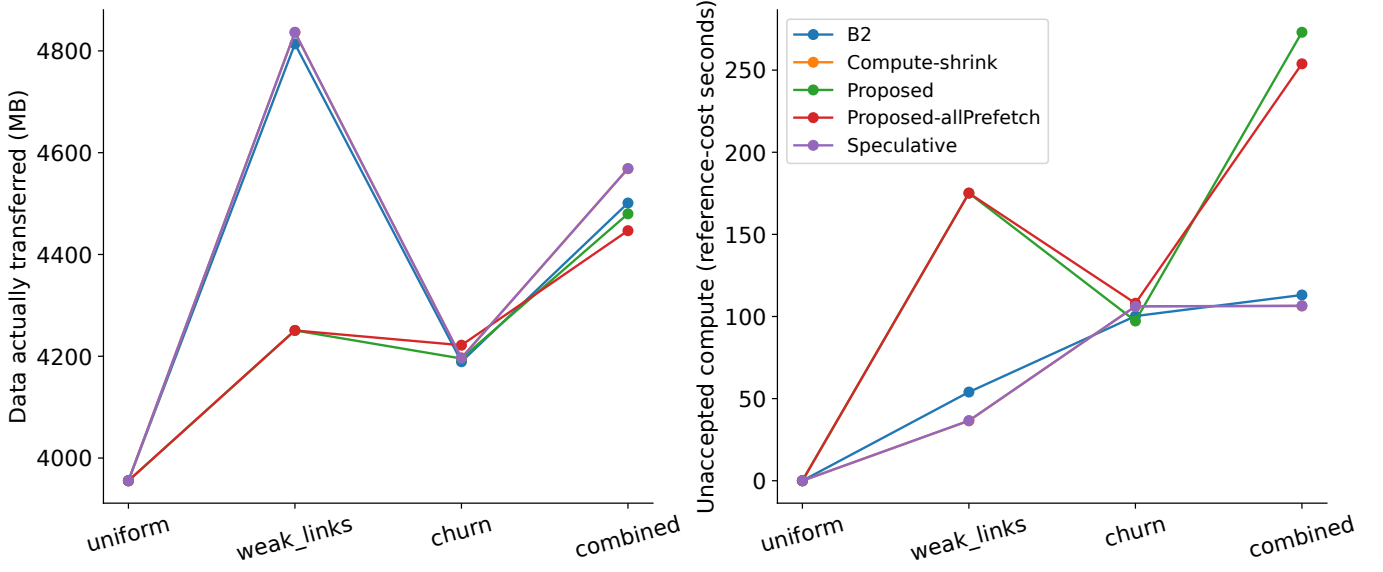


Fig. 7. Data actually transmitted and compute cost not represented by accepted unique tasks. These are simulator counters, not measurements of physical energy or Wi-Fi airtime.

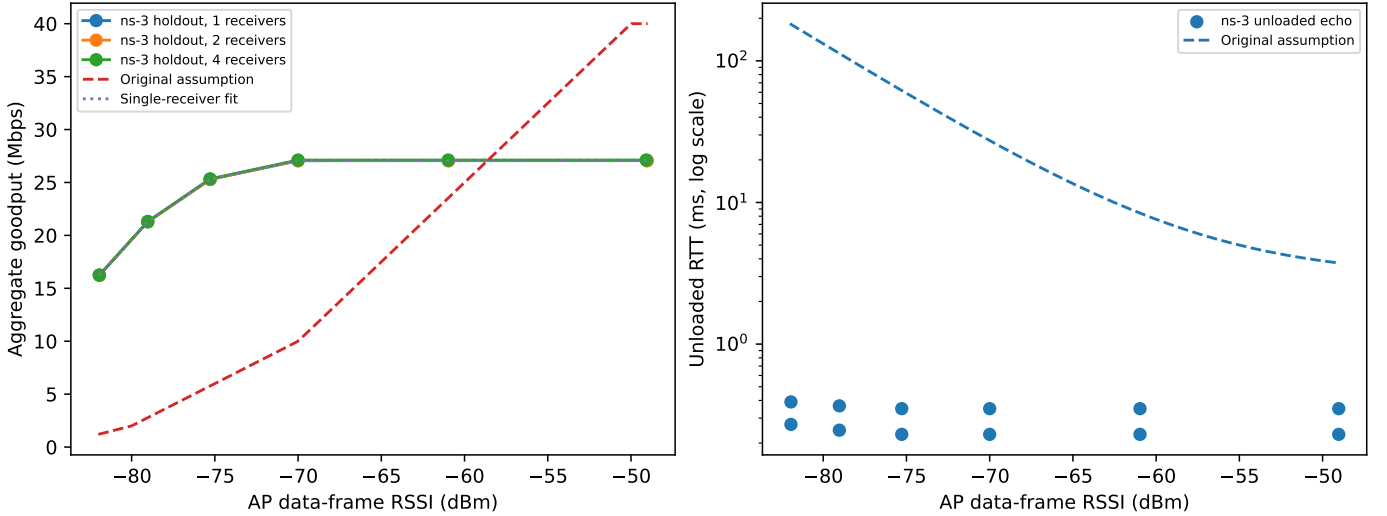


Fig. 8. Independent ns-3 goodput and unloaded-echo diagnostics. The fitted curve uses training seeds only; scheduler evaluations using it remain fluid-channel simulations.

Holm $p = 0.0154$), which meets the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary strong versus HEFT-independent: signed gain 5.16% (paired difference interval [55.87, 73.95] s; global Holm $p = 0.0149$), which meets the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary strong versus Proposed-noRSSI: signed gain 0.00% (paired difference interval [0.00, 0.00] s; global Holm $p = 1.0000$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary strong versus Proposed-noRTT: signed gain 0.00% (paired difference interval [0.00, 0.00] s; global Holm

$p = 1.0000$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary strong versus Proposed-noPrefetch: signed gain 2.05% (paired difference interval [14.88, 34.51] s; global Holm $p = 0.0154$), which meets the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary strong versus Proposed-noCutoff: signed gain 0.34% (paired difference interval [0.62, 8.16] s; global Holm $p = 1.0000$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary mixed versus Tuned-fixed: signed gain 0.28% (paired difference interval [0.04, 7.95] s; global Holm

$p = 1.0000$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary mixed versus Compute-shrink: signed gain 0.75% (paired difference interval [4.70, 16.42] s; global Holm $p = 0.1116$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary mixed versus HEFT-independent: signed gain 5.88% (paired difference interval [62.76, 104.95] s; global Holm $p = 0.0149$), which meets the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary mixed versus Proposed-noRSSI: signed gain 0.00% (paired difference interval [-2.59, 2.43] s; global Holm $p = 1.0000$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary mixed versus Proposed-noRTT: signed gain 0.00% (paired difference interval [0.00, 0.00] s; global Holm $p = 1.0000$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary mixed versus Proposed-noPrefetch: signed gain 0.52% (paired difference interval [1.82, 12.44] s; global Holm $p = 0.9843$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

Ns3 stationary mixed versus Proposed-noCutoff: signed gain -0.06% (paired difference interval [-4.40, 2.80] s; global Holm $p = 1.0000$), which does not meet the prespecified practical-win threshold. The new cells and comparator are exploratory additions.

The mixed-cell gain over the tuned fixed-size baseline fails the practical-win rule. RSSI and RTT removals give identical times in both calibrated cells. Thus, these stationary calibrated experiments do not support an incremental benefit from the link factors. Their high-throughput, low-unloaded-delay regime differs from the assumed weak-link/churn scenarios; it does not disprove benefits under other conditions.

C. Combined-cell censoring sensitivity

A post-hoc extension retains all original combined-cell seeds and the physical trace prefixes while increasing the horizon. These repeated worlds are not independent new evidence. The extended comparisons join the global family; the original horizon comparisons are retained once rather than duplicated.

B2: completion 0.920 versus Proposed 1.000; signed restricted-time gain 72.73% (interval [5391.82, 7196.87] s). Remaining censored runs still prevent an unrestricted mean-completion claim.

B4-F: completion 0.920 versus Proposed 1.000; signed restricted-time gain 72.06% (interval [5168.89, 6994.47] s). Remaining censored runs still prevent an unrestricted mean-completion claim.

Compute-shrink: completion 0.920 versus Proposed 1.000; signed restricted-time gain 71.76% (interval [5127.86,

6864.64] s). Remaining censored runs still prevent an unrestricted mean-completion claim.

Proposed-noRSSI: completion 1.000 versus Proposed 1.000; signed restricted-time gain 33.11% (interval [957.95, 1387.03] s). Remaining censored runs still prevent an unrestricted mean-completion claim.

Proposed-noRTT: completion 1.000 versus Proposed 1.000; signed restricted-time gain 37.98% (interval [1181.51, 1721.90] s). Remaining censored runs still prevent an unrestricted mean-completion claim.

Proposed-allPrefetch: completion 1.000 versus Proposed 1.000; signed restricted-time gain -5.75% (interval [-171.74, -83.68] s). Remaining censored runs still prevent an unrestricted mean-completion claim.

Proposed-noCutoff: completion 1.000 versus Proposed 1.000; signed restricted-time gain -2.84% (interval [-85.40, -44.85] s). Remaining censored runs still prevent an unrestricted mean-completion claim.

D. Published scheduling core comparator

HEFT is evaluated on independent tasks with stationary workers. Upward rank reduces to mean CPU duration across workers; descending ranks are assigned to the worker with earliest predicted CPU finish. Nominal task costs and initial worker speeds are known to this comparator. There are no DAG edges, so communication-edge costs and insertion gaps disappear. Input/output traffic is executed by the shared fluid model after mapping. This is the published CPU-scheduling core for a restricted problem, not a reproduction of LATE, MAESTRO, CROWDio or Honeybee. A hand-calculated ranking/EFT case and end-to-end task-ID test accompany the implementation.

Uniform: HEFT completion 1.000; Proposed signed restricted-time gain 10.46% (interval [87.46, 101.92] s). A censored comparison is not an unrestricted speedup claim.

Weak links: HEFT completion 0.000; Proposed signed restricted-time gain 49.45% (interval [1907.15, 2042.34] s). A censored comparison is not an unrestricted speedup claim.

VIII. DISCUSSION AND LIMITATIONS

The ns-3-derived stationary sensitivity substantially weakens the generality of the original gains: the mixed-link comparison against a tuned fixed-size baseline fails the practical threshold, and RSSI/RTT ablations tie. Neither a realistic low-throughput mobile regime nor coupled packet-level scheduler gains have been established. The combination must justify its incremental value over the matched compute-only policy and the strongest simple alternatives, not only over static allocation. Removing a mechanism may improve the policy. The appendix reports such cases without converting every difference into a win. In the nominal combined cell, unrestricted prefetch also has lower mean makespan, transmitted data and unaccepted compute than stationary-only prefetch. The latter counters are descriptive secondary outcomes, not statistically established advantages. The studied regime therefore does not justify a waste-versus-latency tradeoff explanation for stationary-only prefetch. The completion cutoff is a heuristic:

reporting its comparison with unrestricted continuation is essential, and observing little benefit would argue for simplifying the policy.

The current fixed-size baseline uses a nominated size, and the classical and link-aware families use nominated share parameters. A separate exploratory grid now selects fixed chunk size and weighted share using calibration-only worlds and evaluates the selections on untouched seeds. The paired tuned-baseline results are reported alongside the nominal mechanism study. The matched Compute-shrink comparison keeps the same share and recovery as Proposed, while the separate tuned comparisons check whether this nominal contrast exaggerates improvement over a strong simple baseline. Global correction may render small apparent effects inconclusive; increasing independent seeds is justified only with a recorded precision plan.

The main external-validity limitation is the absence of measured traces and physical end-to-end scheduler validation. An independent ns-3 radio diagnostic now tests a restricted stationary downlink topology, but cannot establish real-device gains or correctness of the mobility model. The code is a new reconstruction from prose and explicit assumptions, rather than the missing original implementation. Reference metadata is verified against Crossref. Full-text novelty review and journal-specific submission instructions remain outstanding. An IEEE journal-format version accompanies this research draft. Source-level tests and analytic channel invariants validate implementation properties but are not substitutes for external wireless validation.

Radio quality is an imperfect departure detector. Default RTT and RSSI are correlated; the additional-delay cell probes a narrower counterexample and does not reproduce all queueing or interference effects. There is a single reflected random-walk mobility family and an operator-controlled stationary designation. No hostile worker, security threat, energy budget, device memory constraint, DAG dependency, joining protocol or master failure is modelled. Claiming measured result correctness from simulation task IDs would be unwarranted.

IX. CONCLUSION

This study implements the requested participation-aware prefetch, completion cutoff, shrinking compute/link-aware chunks and recovery of unfinished task identifiers in SimPy. The key question is where link-aware shrinking helps and where prefetch restrictions hurt. Its data pipeline preserves common worlds, reports completion and truncated makespan together, and produces every reported numerical comparison from raw run files. The calibrated stationary sensitivity fails to establish an advantage over a strong fixed-size baseline. The results define conditional hypotheses for a physical prototype; they do not guarantee superior performance, independent novelty, or journal acceptance. The next decisive evidence is externally measured wireless behaviour and a small repeated scheduler testbed, accompanied by verified prior-art comparisons and calibrated baselines.

DATA, CODE AND AUTHOR STATEMENTS

Code, configurations, raw results, analysis and figure sources are included in the local HPC artefact. The ZIP artefact has been uploaded to the research-deliverables branch of logiqueviper/HPC on GitHub; repository visibility has not been independently established. No archival Zenodo DOI is claimed. Public deposition requires the authors to review the final artefact and its licensing. The listed author and institution are taken from the supplied synopsis; corresponding-author details, coauthor consent and declarations remain to be confirmed. No human-subject measurements or identifiable device traces were collected in this simulation. Any future device study requires participant consent, privacy protection and the institution's applicable review.

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SUPPLEMENTARY MATERIAL

Complete scenario parameters, executable policy switches, per-cell completion and resource counters, all robustness comparisons and permutation-resolution/precision diagnostics accompany this draft in supplement.pdf. The machine-readable claim ledger links each quantitative comparison to its generated macros and raw-run analysis.